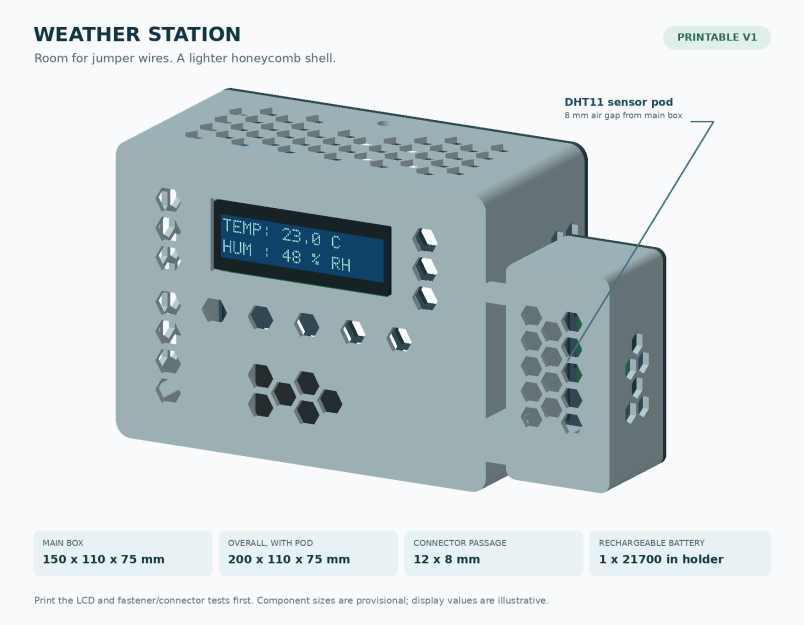
Room to build comfortably

A roomy, ventilated desktop enclosure for a bare 16-pin LCD1602A, DHT11 sensor, ESP32-S2 Mini and one rechargeable 21700 cell.



The first print is a fit test. This is a dimensioned CAD prototype, not a physically assembled or fit-tested product. The actual LCD, battery holder, charger, connectors and switch still determine final fit.

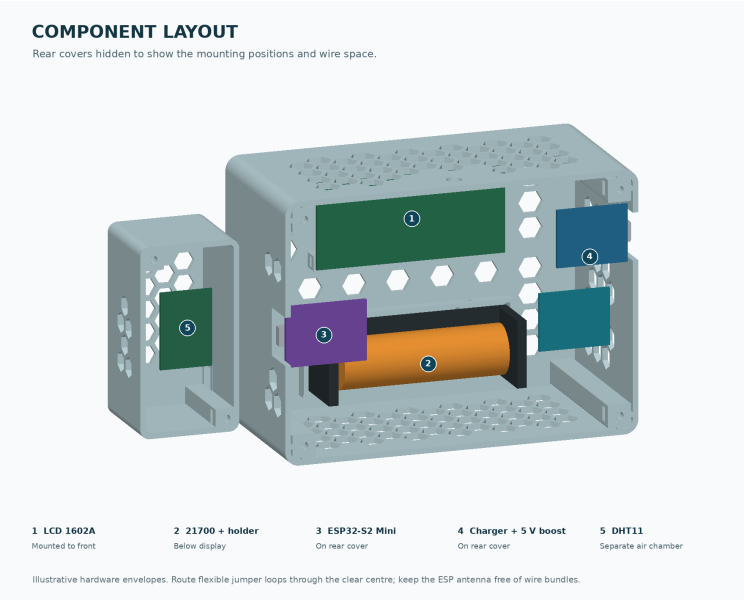
Feature	V1 design
Main enclosure	150 wide x 110 high x 75 deep mm
Sensor pod	42 wide x 80 high x 42 deep mm; 8 mm air gap
Overall assembled envelope	200 wide x 110 high x 75 deep mm, excluding screw heads and external cables
Walls and ventilation	2.4 mm walls; real open honeycomb cutouts with solid frames
Service access	Separate removable rear covers; generous central space for jumper wires

The DHT11 sits in its own ventilated side pod. The air gap reduces direct heat conduction from the display and electronics, but it does not guarantee accurate temperature readings while the charger or ESP32 is warm.

Use: dry indoor placement, shaded from direct sun. The open vents are not rainproof or splashproof.

Inside the enclosure

Illustrated electronics show size envelopes. Only the three plastic enclosure parts and two fit coupons belong in the print files.



Component	Assumed fit / mounting
Bare LCD1602A	80 x 36 mm PCB; 75 x 31 mm hole centres; 73.2 x 27.2 mm front opening. Four M2 mounts. No I2C backpack is assumed.
ESP32-S2 Mini	34.3 x 25.4 mm board. Rear-cover support pads and strap slots; allow space for your fitted headers and connectors.
DHT11 module	Reserved envelope: up to 26 x 38 x 22 mm, including protrusions. Strap the breakout board; keep the sensing grille open.
21700 holder	Commercial single-cell holder up to 84 x 28 x 25 mm; tray space 86 x 30 mm. Use the holder contacts, not printed battery contacts.
Power modules	Two generic rear-cover zones, each 30 x 24 mm. Actual modules, connector direction and component heights need checking.

Access openings: two 20 mm-wide USB slots open to the rear edge, extending 16.1 mm into the shell; 6.5 mm switch hole for a provisional M6 threaded toggle. The open slots let a populated rear cover slide into place. Actual hardware alignment still needs checking.

Wire route: a 12 x 8 mm passage links the main chamber to the sensor pod. Leave a service loop so the covers can be opened without pulling on pins.

Parts and print setup

Print the two small tests before committing material to the main enclosure. All STL files are already oriented with the intended bed face down.

Printed part	Quantity / purpose
01_LCD_Fit_Test.stl	1; check opening, mounting holes and LCD depth
02_Fastener_Fit_Test.stl	1; check M2/M3 nuts, bolt passages and 12 x 8 mm sensor connector passage
03_Main_Shell_With_Sensor_Pod.stl	1; front face on the bed, rear openings upward
04_Main_Rear_Cover.stl	1; outside face on bed, inside supports upward
05_Sensor_Pod_Rear_Cover.stl	1; outside face on bed, inside supports upward

Hardware and mounting supplies

Item	Quantity / selection
Rear cover fasteners	6 x M3 x 10 mm machine screws + 6 x M3 hex nuts
LCD fasteners	4 x M2 x 6 mm machine screws + 4 x M2 hex nuts; confirm they pass through your LCD holes
Board / holder retention	Small cable ties or reusable straps; thin nonconductive foam pads where needed
Electronics	LCD1602A, DHT11, ESP32-S2 Mini, 1 x 21700 holder and cell, suitable 1S charger/protection, regulated 5 V boost, power switch

Suggested starting settings

For an indoor prototype, start with PLA, a 0.4 mm nozzle, 0.20 mm layers, 3 walls, 4 top/bottom layers and 15% infill. The hexagons are holes in the model; changing infill does not create or close them.

Review bridges and overhangs in your slicer. The shell includes side openings, hexagons and a short cable-duct roof; supports may be useful in selected areas depending on your printer. Keep removable support away from nut pockets where possible.

The three enclosure parts contain 192.3 cm3 of CAD solid volume. The extra honeycomb removes 14.0% relative to the same design without those extra openings. This comparison excludes fit tests and is not a slicer filament estimate; the sensor vents remain in both versions.

3MF: geometry only, with all five parts arranged in a roughly 254 x 232 mm area. It contains no printer profile, filament profile, slicing settings or G-code. Use the STL tests separately first.

Assembly, step by step

Work with the battery removed and USB cables unplugged until the mechanical fitting is complete.

1. Test the LCD and fasteners. Place your actual LCD behind the test frame. Check the four mounting holes, opening and front-to-PCB distance. Test the nuts and bolts without forcing them. Pass the actual sensor connector through the 12 x 8 mm test tunnel.

2. Print and clean the enclosure. Remove loose strands from honeycomb cells, bolt holes, nut slots and the cable passage. Check that each cover seats without pressure. If a fit test fails, adjust the CAD before printing the larger parts.

3. Load the captured nuts. Slide four M2 nuts into the LCD mounts and six M3 nuts into the cover posts. The pockets load from the side. Keep nuts fully seated and parallel to the screw axis.

4. Install the display. Seat the LCD on its four supports. Secure it with four M2 x 6 mm screws, gently and evenly. The screws hold the PCB; they must not press the glass or require bending the board.

5. Secure the battery holder. Route two small ties through the underfloor tunnels in the tray and around the commercial holder. Check holder fit and wire exit before inserting the cell. Do not solder directly to the lithium cell.

6. Mount the rear electronics. Fit the ESP32, charger and boost boards to the rear-cover support zones with nonconductive pads and ties. Keep board undersides insulated and components unobstructed. Adjust board height so each USB socket aligns with its access opening.

7. Fit the sensor and route the jumpers. Secure the DHT11 breakout to the pod cover using its strap slots. Do not cover the blue sensing grille. Route its cable through the 12 x 8 mm bridge and allow slack for removing the cover.

8. Check, test and close. Check power polarity, connector orientation and voltage before connecting the ESP32 or LCD. Route jumpers away from screw posts and the ESP32 antenna. Test with covers open, then close using the six M3 screws without trapping wires.

Make assembly easier

Use pre-crimped Dupont leads once the required pin headers are fitted. This model deliberately leaves a deep central chamber for plug housings, wire bends and a service loop. The bare parallel LCD still needs its control/data wiring and contrast/backlight circuit; the enclosure does not replace those connections.

Before final closure: check that the LCD contrast control remains reachable, the switch moves freely, both USB plugs enter without pushing the PCB, and no wire touches an exposed battery terminal.

Power, limits and references

The case reserves space for a power system. Charging and USB power routing must match the actual modules you install.

A practical single-cell power arrangement

Use **one 21700 lithium-ion cell** in a commercial holder, a suitable **1S charger and protection circuit**, and a **regulated 5 V boost converter** for the station. Choose a boost output that can handle the actual display load and ESP32 Wi-Fi current peaks. Verify its voltage before attaching either board.

The ESP32-S2 Mini USB connector powers/programs the board; it is not a battery charger. An XL4005 is a step-down converter, so it cannot raise one lithium cell to a regulated 5 V. The power-module drawing is illustrative, not an electrical schematic.

Charging: with a basic TP4056 setup that lacks proper load sharing, switch the station off while charging. The power switch should disconnect the boost 5 V supply from the station. Disconnect that battery-derived supply before connecting the ESP32 to a computer USB port; do not join two 5 V sources without a verified power-selection circuit.

Signal levels: the ESP32 uses 3.3 V logic. A 5 V LCD or sensor board does not automatically guarantee compatible signal levels. Confirm the exact modules before choosing the connection circuit; the pin-by-pin wiring depends on those parts.

Verification boundary: CAD validation is recorded in geometry_report.json. Printed fit, actual USB alignment, hardware clearances, electrical compatibility, slicing, runtime and temperature accuracy still need real-world checks.

Primary references

WEMOS - S2 Mini specifications and schematic links
https://www.wemos.cc/en/latest/s2/s2_mini.html

Winstar - representative 80 x 36 mm LCD mechanical drawing
https://www.winstar.com.tw/uploads/editor/files/WH1602B-16x2-datasheet.pdf

Aosong - DHT11 manufacturer datasheet, distributor-hosted copy
https://www.haoyuelectronics.com/Attachment/DHT11/DHT11.pdf

Aosong - DHT11 enclosure and heat-source guidance, page 12
https://static.maritex.eu/file/display/jOwQ3b4egHyea8BUoPT1wkbTd5WP0Pb0/DHT11.pdf

Espressif - ESP32-S2 antenna and enclosure guidance
https://docs.espressif.com/projects/esp-hardware-design-guidelines/en/latest/esp32s2/pcb-layout-design.html

WEMOS - S2 Mini schematic, USB VBUS / 5 V rail
https://www.wemos.cc/en/latest/_static/files/sch_s2_mini_v1.0.0.pdf

TP4056 - single-cell charger datasheet
https://dlnmh9ip6v2uc.cloudfront.net/datasheets/Prototyping/TP4056.pdf

Microchip AN1149 - lithium battery load sharing
https://ww1.microchip.com/downloads/en/AppNotes/01149a.pdf

XLSEMI - XL4005 step-down converter datasheet
https://www.xlsemi.com/datasheet/XL4005-EN.pdf

The Winstar LCD is a dimensional reference, not identification of your exact module. Model variants differ. Sources checked 24 October 2026; all design choices and clearances remain prototype assumptions.